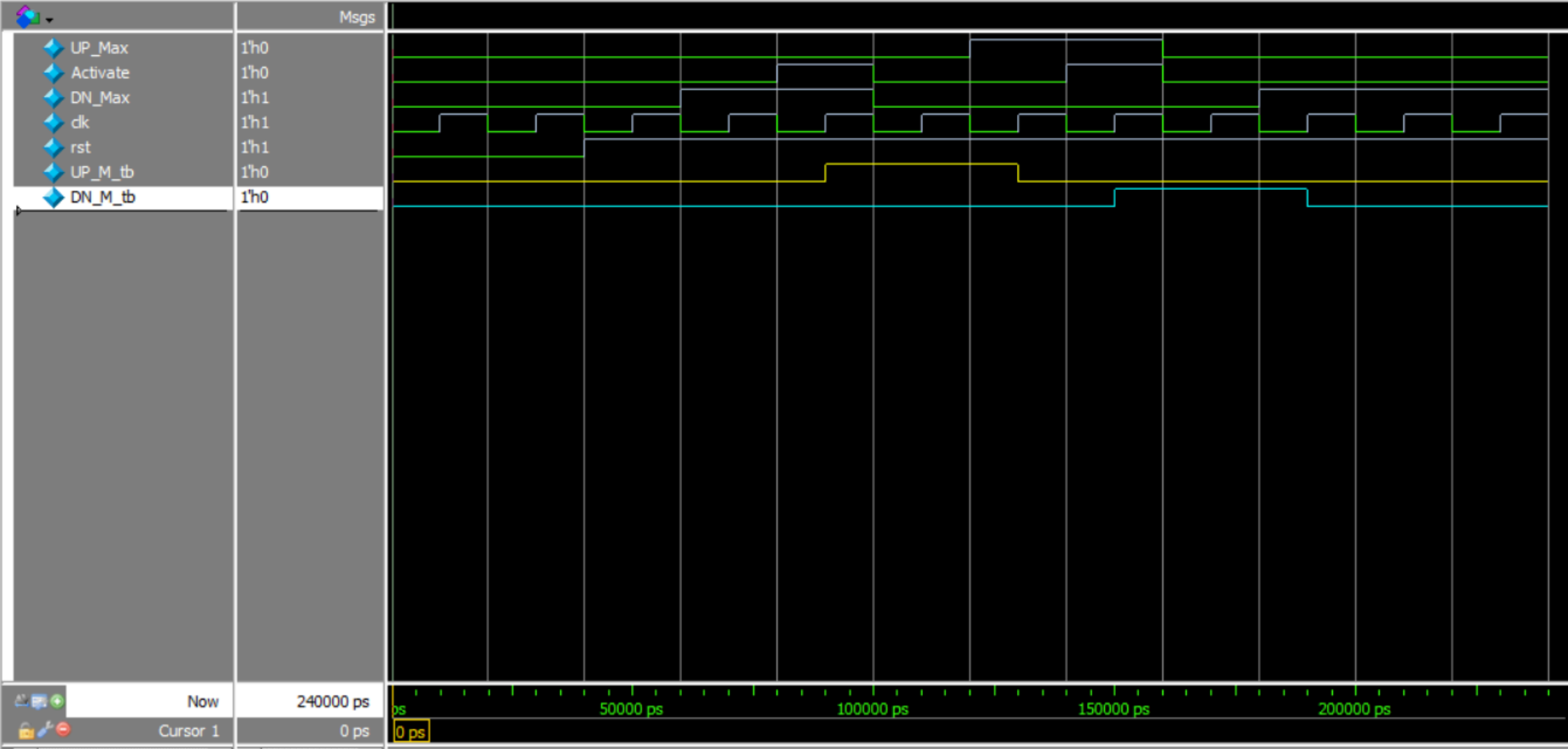

Msgs
UP_Max
Activate
DN_Max
clk
rst
UP_M_tb
DN_M_tb
1'h0
1'h0
1'h1
1'h1
1'h1
1'h0
1'h0
Now
240000 ps
Cursor 1
0 ps
50000 ps
100000 ps
150000 ps
200000 ps
0 ps

```
# Top level modules:
#       Controller_TB
# End time: 00:10:59 on Jul 22,2026, Elapsed time: 0:00:00
# Errors: 0, Warnings: 0
# vsim -voptargs="+acc" work.Controller_TB -coverage -sv_seed random -l siml.log
# Start time: 00:10:59 on Jul 22,2026
# ** Note: (vsim-3812) Design is being optimized...
# ** Note: (vopt-143) Recognized 1 FSM in module "Controller(fast)".
# Loading work.Controller_TB(fast)
# Loading work.Controller(fast)
# Sv_Seed = 1249699681
# [PASS]                                                    1. Initial IDLE State Check | UP_M: 0 | DN_M: 0 time 80000
# [PASS]                                                    2. IDLE -> Mv_UP Transition | UP_M: 1 | DN_M: 0 time 100000
# [PASS]                                                 3. Mv_UP Hold (Door moving up) | UP_M: 1 | DN_M: 0 time 120000
# [PASS]                                                    4. Mv_UP -> IDLE Transition | UP_M: 0 | DN_M: 0 time 140000
# [PASS]                                                    5. IDLE -> Mv_Dn Transition | UP_M: 0 | DN_M: 1 time 160000
# [PASS]                                               6. Mv_Dn Hold (Door moving down) | UP_M: 0 | DN_M: 1 time 180000
# [PASS]                                                    7. Mv_Dn -> IDLE Transition | UP_M: 0 | DN_M: 0 time 200000
# ALL TESTS COMPLETED.
# ** Note: $finish    : Controller_TB.v(95)
#    Time: 240 ns  Iteration: 0  Instance: /Controller_TB
# 1
# Break in Module Controller_TB at Controller_TB.v line 95
```